QUANT

Percentages and Profit & Loss

Assignment 05

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Percentage, P & L Online Assignment 05

1. A trader bought 80 articles. He sold 40 % of them at a loss of Rs. 600 and the rest for Rs. 6,000 and gained 30 % on the total outlay. What was the cost of each article?

1. Rs. 75 2. Rs. 100 3. Rs. 50 4. Rs. 37.50 5. Rs. 90

2. A CD player manufacturer marks his product 87.5 % above its cost price. He allows his distributor a discount of 40 % on the list price and the distributor allows the retailer a discount of 30 % on the marked price. Find the profit margin of the manufacturer and the retailer.

1. 12.5 %, 42.86 % 2. 12.5 %, 16.67 % 3. 42.86 %, 16.67 %
4. 12.5%, 25% 5. 16.67%, 33.33%

3. A man sells an article at a profit of 5 per cent. If the cost price were 5 per cent less and the selling price Rs. 1 less, his profit would have been 10 per cent. Find the cost price of the article.

1. Rs. 100 2. Rs. 400 3. Rs. 225 4. Rs. 180 5. Rs. 200

4. A Bombay merchant has to remit some money to England. A moneychanger takes 5 paisa more than the bank rate per rupee per pound. The merchant reckons that he needs Rs. 3,660, if he converts his money through the changer, while he needs Rs. 3650 through the bank for sending the same amount of English money. What is the bank rate of exchange in Rupees/ pound and how much money had he to remit in pound sterling?

1. Rs. 17.25, pounds 200 2. Rs. 18.25, pounds 200
3. Rs. 17.25, pounds 300 4. Rs. 18.25, pounds 300
5. Rs. 19.25, pounds 250

5. A dishonest dealer professes to sell his goods at a profit of 20 percent and also weighs 800 grams in place of a kg. Find his actual gain percent.

1. 45 % gain 2. 50 % gain 3. 40 % gain
4. No gain no loss 5. 60% gain

6. A manufacturer makes an article at a cost of Rs. 250. He catalogues it at 60 % above the cost price and allows a trade discount of 12 ½ % off the catalogue price and a cash discount of 12½% off the price so reduced. What profit does he make on each article?

1. Rs. 56.25 2. Rs. 112.50 3. Rs. 50 4. Rs. 250 5. Rs. 75

7. A tradesman sells his goods at a price 30 % above what they cost him. If his overhead expenses are Rs. 57,800 per annum, what must be his annual receipts on the goods sold, so that 10 % of his receipts may be net profit?

1. Rs. 4,00,000 2. Rs. 3,50,000 3. Rs. 4,42,000 4. Rs. 8,84,000 5. Rs. 2,21,000

8. The population of the village of Dehlan is 10,000 at this moment. It increases by 10 % in the first year. In the second year, due to immigration, the population drops by 5 %. Find the population at the end of the third year if in the third year the population increases by 20 %.

1. 12,340 2. 12,540 3. 1,27,540 4. 12,340 5. 1,25,400

9. In an assembly election at Surat, the total turnout was 80 %, out of which 16 % of the total voters on the voting list were declared invalid. Find which of the following can be the percentage votes got by the winner of the election, if the candidate who came second got 20 % of the total voters on the voting list. (There were only three contestants, only one winner and the total number of voters on the voters' list was 20,000.)

1. 44.8 % 2. 46.6 % 3. 48 % 4. 28% 5. 46%

10. The population of a city is 200,000. If the annual birth rate and the annual death rates are 6 % and 3 % respectively, then calculate the population of the city after 2 years.

1. 212,090 2. 206,090 3. 212,000 4. 212,180 5. 2,12,280

Percentage, P & L Assignment 05		
1.	1	Let the CP of each article = x. SP of 80 articles to gain 30% on the whole $= 80x \times \frac{130}{100}$. He sold 32 at a loss of 600 and rest for 6000. $\therefore 80x \times \frac{130}{100} = (32x - 600) + 6000 \Rightarrow x = 75$.
2.	1	Let CP of manufacturer = 100 $\therefore$ his MP = 187.5. SP of manufacturer = $187.5 \times \frac{60}{100} = 112.5$ $\therefore$ profit margin of the manufacturer = $\frac{112.5-100}{100} \times 100 = 12.5\%$. C.P of distributor = 112.5. Now SP of distributor = $\frac{70}{100} \times 187.5 = 131.25$. $\therefore$ profit margin of distributor = $\frac{131.25-112.5}{112.5} \times 100 = 16.67\%$. CP of retailer = 131.25. SP of retailer = 187.5 $\therefore$ profit margin of retailer = $\frac{187.5-131.25}{131.25} \times 100 = 42.86\%$.
3.	5	Let CP of the article = 100 $\therefore$ Old SP = 105. New CP = 95. New SP = $95 \times \frac{110}{100} = 104.5$. If difference in SP is 0.5 then CP = 100, If difference in SP is 1 then CP = $\frac{100}{0.5} \times 1 = 200$.
4.	2	Moneychanger takes 5 paisa more per unit of currency. If difference is 5p, he is to remit 1 unit. If the difference is 1000p, he is to remit 1/5 * 1000 = 200 units i.e. pounds 200. Also bank rate of exchange = $\frac{3650}{200} = Rs.18.25$.
5.	2	Uses 800 gms instead of 1000 gms $\therefore$ % profit = $\frac{1000-800}{800} \times 100 = 25\%$. So Net profit = $1.2 \times 1.25 = 1.5 \Rightarrow 50$ %.
6.	1	CP = 250. MP = $\frac{160}{100} \times 250 = 400$ $\therefore$ SP = $400 \times \frac{87.5}{100} \times \frac{87.5}{100} = 306.25$ $\therefore$ Profit = 306.25 – 250 = Rs. 56.25.
7.	3	Let CP = 100 $\therefore$ SP = 130. 10% of 130 = $130 \times \frac{10}{100} = 13$ $\therefore$ Real CP = 130 – 13 = 117. $\therefore$ Overheads = 117 – 100 = 17. If overheads is 17 then SP = 130 $\therefore$ if overheads is 57800, then SP = $\frac{130}{17} \times 57800 = 442000$.
8.	2	If the original population is 1, the new population will be $1.1 \times 0.95 \times 1.2 = 1.254$. Thus 10000 will become 12540.
9.	4	Total turnover was 80%, out of which 16% of total was invalid. Thus valid votes are 64% of

		the total. The second candidate got 20% the remaining votes are 44% of the total. Now the winner must have got more than 20% as he is the winner and less than 44%, which is the remaining votes. So the answer is 28% which is 4th option.
10.	4	Required value = $200000 \times (1.03)^2 = 212180$.